(V) Sliding Window Approach:

- Scans entire image with different window sizes
- Detects objects at different positions & scales

→ Example:

Haar Cascade is commonly used for face detection. In OpenCV, a pre-trained model like haarcascade_frontalface_default.xml is used to detect faces in images or live camera feed.

Q.) Object Detection & Common Techniques.

→ Object Detection is a computer vision technique used to identify & locate objects in an image or video.

- It ~~not~~ only classifies objects but also draws bounding boxes around them

⇒ Key Points:

(I) Identification & Localization:

- Detects what the object is (classification)
- Find where the object is (location using bounding box)

(II) Bounding Box:

- Rectangular boxes drawn around detected objects
- Shows position & size of object.

Q) Explain about Haar Cascade Technique.

→ Haar Cascade Technique is a machine learning based object detection method used to detect object like faces, eyes & cars in images or video.

- It uses Haar-like features & a cascade classifier to perform fast detection.

⇒ Key Points:

1) Haar like Features:

- Uses black & white rectangular patterns.
- Detects difference in pixel intensity
- Example: eyes are darker than cheeks.

(II) Integral Image:

- Used to speed up calculation
- Makes detection fast

(III) AdaBoost Algorithm:

- Select important features
- Remove unnecessary features

(IV) Cascade Classifier:

- Detection happens in multiple stages
- Non-object regions are rejected early
- Improves speed & efficiency.

Q.) What is RANSAC? How does it work? Why to use RANSAC?

⇒ RANSAC (Random Sample Consensus) is an algorithm used to estimate a mathematical model from data that contains noise & outliers.
- It helps in finding the best fit by ignoring incorrect data points.

⇒ Key Points:—

1) Handles Outliers:
- works even if data contains wrong or noisy points
- ignores incorrect data.

II) Robust Algorithm:
- Gives accurate result even with imperfect data.
- Better than normal fitting ~~methods~~ methods.

III) Iterative Process:
- Repeats steps multiple times
- Selects best model based on maximum (correct points).

⇒ How It works:—

(i) Randomly select a small subset of data
(ii) Fit a model using that subset
(iii) Check how many data points fit the model (inliers)
(iv) Repeat the process many times
(v). Choose the model with the maximum inliers

(III) Multiple Object Detection:
- Can detect more than one object in a single image.
- Example: Person / car / bike together.

(IV) Confidence Score:
- Each detected object has probability score
- Shows how accurate the detection is

⇒ Common Technique Used:

1) Traditional Techniques
- Haar Cascade

2) Machine Learning Techniques
- SVM (Support Vector Machine)
- classifies objects based on features

3) Deep Learning Techniques (Modern)
- CNN (Convolutional Neural Network)
- Automatically learns features.

⇒ Example:
- In a traffic image, object detection can identify & locate cars, buses, & pedestrians by drawing bounding boxes around them

⇒ Why to Use RANSAC:

- Real world data has noise & errors
- Normal methods fail with outliers
- RANSAC gives more reliable results

⇒ Example;

RanSAC is used in image stiching to find matching points between two images & remove incorrect matches (outliers)